# Sahaj Bhadja

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## Summary

Full Stack Software Engineer with 2+ years of experience shipping **TypeScript/React products**, **Node.js/Python services**, LLM retrieval workflows, tested APIs, and **reliable education software** across production environments.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems

## Experience

**USC Dornsife** | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Reduced dashboard latency from 1.2s to 350ms through **React interfaces, SQL indexing, and query optimization**, supporting 500+ users with responsive, maintainable analytics workflows across isolated tenant environments.
- Achieved 99%+ reliability across 5K+ monthly jobs through **validation, reconciliation, and secure error handling**, cloud storage, CI/CD releases, and production monitoring for dependable data-intensive software workflows at scale.

**Grad8** | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, secure ingestion, AWS RDS, and CI/CD workflows.
- Processed 1K+ files per batch using **concurrent Python services**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during sustained production traffic and model execution reliably.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* | *India*
- Improved API response times by 30% through **maintainable microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments under sustained traffic.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

**Aasma Technology Solutions** **May 2023 – August 2023**
*Full Stack Developer Intern - (VibrantChem)* | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform**, designing reusable interfaces for customers, billing, allocation, permissions, and notifications with real-time updates across distributed locations reliably.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT access controls, WebSocket updates, Azure monitoring, automated tests, and Docker/Jenkins deployments across eight modular service domains.

**LST Capitals** | *Web Developer Intern* | *India* **November 2022 – January 2023**
- Improved cross-platform compatibility by 15% through **React Native and TypeScript interfaces**, integrating Android, iOS, and web booking workflows while reducing design errors by 25% through reusable components.
- Delivered **sub-200ms customer workflows** across 50K+ booking records, integrating pricing, coupons, payment webhooks, Firebase synchronization, and status updates while preventing conflicting reservations across concurrent user sessions reliably.

## Skills

**Languages:** TypeScript, JavaScript, Python, Java, SQL, C++, Bash
**Frontend:** React, Next.js, React Native, HTML, CSS, Tailwind CSS, Responsive Design, Real-Time Interfaces
**Backend & APIs:** Node.js, FastAPI, REST APIs, WebSockets, Microservices, JWT, Background Processing, Third-Party Integrations
**AI & Retrieval:** LLaMA-2, FAISS, Pinecone, Semantic Search, Vector Search, Embeddings, Transformers, Model Evaluation
**Data & Cloud:** PostgreSQL, MongoDB, Data Modeling, Schema Design, Query Optimization, AWS, GCP, Azure
**Engineering:** Data Structures, Algorithms, Object-Oriented Programming, System Design, Git, GitHub, CI/CD, Unit Testing, Integration Testing, Automated Testing, Debugging, Production Reliability

## Projects

**Big-Brain** | Live
- Built a **secure AI document platform** with LLaMA-2, FAISS, Pinecone, PostgreSQL, semantic retrieval, role-based access, and real-time collaboration, supporting 50K+ documents while reducing analysis time by 60%.

**Chat2SQL** | Live
- Extended a **Java and TypeScript AI database workspace** supporting 10+ database systems, using Docker, Maven, Node.js, and Yarn for schema exploration, query workflows, validation, debugging, and repeatable deployment.

**HMIS - Hospital Management Information System** | Live
- Built a **TypeScript operations platform** integrating consent-based APIs, Twilio alerts, and Sentry monitoring, enabling 100+ daily appointments, 2x-faster record access, and 30% lower administrative workload securely for users.